

Point

From Number Theory to Algebraic Geometry and Logic

Session P0 — Set Theory, Order, and the Axiom of Choice

Complete lecture notes with proofs, examples, and exercises

Abstract

This is the true session zero of the course — the one the first class actually needed. It builds the set-theoretic scaffolding everything else stands on: equivalence relations and their exact correspondence with partitions (the logic of every quotient to come: $\mathbb{Z}/n$, G/N , $M \otimes N$, sheafification); order relations (preorder, partial order, strict order) and the anatomy of extrema (minimum vs minimal, bounds, well-orders); and the axiomatic frame ZF with the **Axiom of Choice**, its equivalents (Well-Ordering, Zorn’s lemma — with the equivalence cycle proved or sketched), and its famous consequences and equivalents: bases of vector spaces, Tychonoff, Tarski’s theorem, spanning trees, and the precise tier of the Completeness Theorem. Zorn’s lemma is the engine of three existence theorems later in the course: Krull (S1), algebraic closures (S3), and Deligne (S58).

Session plan (150 minutes)

Time	Block
0:00–0:10	Orientation: why set theory first; where P0 feeds (S1, S3, S2, S27, S58)
0:10–0:35	§1 Equivalence relations and partitions (the logic of quotients)
0:35–1:00	§2–§3 Orders: preorder/partial/strict; minima vs minimal; bounds; well-orders
1:00–1:10	<i>Break</i>
1:10–1:35	§4 From Russell’s paradox to ZF
1:35–2:10	§5 Axiom of Choice; the equivalence cycle; the list of equivalents
2:10–2:30	Exercises / wrap-up: Zorn as the course’s existence engine

Conventions

A *relation* on X is a subset of $X \times X$; we write $x \sim y$ for $(x, y) \in R$. Orders are written $\preceq$ (reflexive form) and $\prec$ (strict form). “Poset” means partially ordered set. ZF denotes Zermelo–Fraenkel set theory without Choice; ZFC = ZF + AC.

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1 Equivalence Relations and Partitions

Definition 1.1 (Equivalence relation; class; quotient). A relation $\sim$ on X is an **equivalence relation** if it is reflexive ($x \sim x$), symmetric ($x \sim y \Rightarrow y \sim x$), and transitive. The **class** of x is $[x] = \{y \in X : y \sim x\}$; the **quotient set** is $X/\sim = \{[x] : x \in X\}$.

Definition 1.2 (Partition). A **partition** of X is a family $\mathcal{P} \subseteq \mathcal{P}(X)$ of nonempty, pairwise disjoint sets whose union is X .

Theorem 1.3 (Equivalence $\leftrightarrow$ partition). (a) *If $\sim$ is an equivalence relation on X , then $X/\sim$ is a partition of X .*

(b) *If $\mathcal{P}$ is a partition of X , then $x \sim_{\mathcal{P}} y \iff \exists P \in \mathcal{P}, x, y \in P$ is an equivalence relation, and its classes are exactly the members of $\mathcal{P}$.*

Hence equivalence relations on X and partitions of X are in canonical bijection.

Proof. (a) Each $x \in [x]$ by reflexivity, so the union is X and classes are nonempty. If $z \in [x] \cap [y]$ then $x \sim z$ and $z \sim y$, so $x \sim y$; then for any $u \in [x]$: $u \sim x \sim y$ gives $u \in [y]$, hence $[x] \subseteq [y]$; symmetrically $[y] \subseteq [x]$. So distinct classes are disjoint. (b) Reflexive since $\mathcal{P}$ covers X ; symmetric by definition; transitive because if $x, y \in P$ and $y, z \in Q$ then $y \in P \cap Q$, forcing $P = Q$ by disjointness, so $x, z \in P$. The class of x is the unique $P \ni x$. The two passages invert each other. $\square$

Example 1.4 (Quotients everywhere). • Congruence mod n on $\mathbb{Z}$: classes $[0], \dots, [n-1]$; quotient $\mathbb{Z}/n\mathbb{Z}$ (S1).

- Construction of $\mathbb{Q}$: on $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, $(a, b) \sim (c, d) \iff ad = bc$; checking this is an equivalence relation *is* the construction of the rationals.
- Cosets G/H (P1), orbits of a group action (P1), stalks and sheafification (S26–S27), tensor products as quotients of free modules (S2): every quotient in the course is Theorem 1.3 at work — *identifying what should be the same point*.

Remark 1.5 (Choosing representatives). A quotient map $X \rightarrow X/\sim$ generally has no canonical section: picking one representative per class for infinitely many classes is exactly where the Axiom of Choice (§5) enters. Keep this in mind for S2 and S27.

2 Order Relations and Their Kin

Definition 2.1 (Preorder, partial order, strict order, total order). A relation $\preceq$ on P is a **preorder** if reflexive and transitive; a **partial order** if also antisymmetric ($x \preceq y \preceq x \Rightarrow x = y$); **total/linear** if any two elements are comparable. A relation $\prec$ is a **strict partial order** if irreflexive and transitive; a **strict total order** if additionally $x \prec y$ or $x = y$ or $y \prec x$ for all x, y .

Proposition 2.2 (The two forms of order correspond). *If $\preceq$ is a partial order then $x \prec y \iff (x \preceq y \wedge x \neq y)$ is a strict partial order; if $\prec$ is a strict partial order then $x \preceq y \iff (x \prec y \vee x = y)$ is a partial order. These passages are mutually inverse.*

Proof. Irreflexivity of $\prec$ is immediate; transitivity: if $x \prec y \prec z$ then $x \preceq z$, and $x = z$ would give $x \preceq y \preceq x$, hence $x = y$ by antisymmetry, contradicting $x \prec y$; so $x \neq z$, i.e. $x \prec z$. The converse direction and the round-trips are similar case analyses. $\square$

Proposition 2.3 (A preorder is a partial order on its quotient). *If $\preceq$ is a preorder on P , then $x \sim y \iff (x \preceq y \wedge y \preceq x)$ is an equivalence relation, and $[x] \preceq [y] \iff x \preceq y$ is a well-defined partial order on $P/\sim$. Thus §1 and §2 meet: every preorder becomes a poset after quotienting.*

Proof. Reflexivity/symmetry of $\sim$ are built in; transitivity uses transitivity of $\preceq$ twice. Well-definedness: if $x \sim x'$ and $x \preceq y$ then $x' \preceq x \preceq y \preceq y'$, so $x' \preceq y'$. Antisymmetry on classes is exactly the definition of $\sim$. $\square$

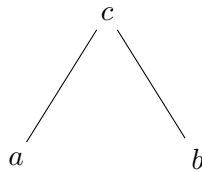
Example 2.4. • $(\mathbb{N}, |)$ divisibility: partial, not total ($2 \nmid 3$, $3 \nmid 2$).

- $(\mathcal{P}(X), \subseteq)$: partial; chains = nested families, antichains = pairwise disjoint (or incomparable) families.
- Preorder of functions $f \preceq g \iff f = O(g)$ (analysis): not antisymmetric — quotienting gives the poset of growth rates (Proposition 2.3).
- Lexicographic order on $\mathbb{R}^2$: total.

3 Extrema, Bounds, and Well-Orders

Definition 3.1 (Bounds and extrema). In a poset $(P, \preceq)$ with $S \subseteq P$: u is an **upper bound** of S if $s \preceq u$ for all $s \in S$ (lower bound dually); the **supremum** is the least upper bound. $m \in S$ is the **minimum** of S if $m \preceq s$ for all $s \in S$; m is **minimal** if no $s \in S$ satisfies $s \prec m$. Maximum/maximal are dual.

Example 3.2 (Minimum $\neq$ minimal). In $P = \{a, b, c\}$ with $a \prec c$, $b \prec c$ and a, b incomparable (Figure 1): the minimal elements are a and b (two of them!), there is *no* minimum; c is both maximum and maximal; the set $\{a, b\}$ has upper bound c (its supremum) but no maximum. In $(\mathbb{N}_{\geq 2}, |)$: minimal elements are the primes; the minimum is 2? no — 2 does not divide 3: there is no minimum, only minima.



two minimal elements, no minimum; $c = \text{maximum} = \text{supremum of } \{a, b\}$

Figure 1: Hasse diagram: minimal vs minimum, maximal vs maximum, bounds vs extrema.

Definition 3.3 (Well-order). A total order $\preceq$ on W is a **well-order** if every nonempty $S \subseteq W$ has a minimum. Examples: $(\mathbb{N}, \leq)$; $(\mathbb{Z}, \leq)$ and $(\mathbb{R}, \leq)$ are *not*; every ordinal is.

Proposition 3.4. *A well-order has no infinite strictly decreasing chain $x_1 \succ x_2 \succ \dots$; conversely, a total order with no infinite decreasing chain is a well-order provided countable choice is available (over ZF the forward direction is unconditional).*

Proof. If $x_1 \succ x_2 \succ \dots$ exists then $\{x_i : i \in \mathbb{N}\}$ is nonempty with no minimum. Conversely, if $S \neq \emptyset$ has no minimum, pick $x_1 \in S$; having x_n , the set $\{s \in S : s \prec x_n\}$ is nonempty (else x_n were minimal=minimum), so choose x_{n+1} in it — producing a decreasing chain. (The recursive choices use dependent choice; note where.) $\square$

4 From Russell's Paradox to ZF

Proposition 4.1 (Russell). *There is no set R with $R = \{x : x \notin x\}$.*

Proof. If R were a set, then $R \in R \iff R \notin R$ by definition — contradiction. Hence “the set of all sets” cannot exist: comprehension must be restricted. $\square$

Definition 4.2 (The ZF axioms, informally). Language: only $\in$ and $=$. Axioms: **Extensionality** (same elements $\Rightarrow$ same set); **Pairing** $\{x, y\}$; **Union** $\bigcup \mathcal{F}$; **Power set** $\mathcal{P}(X)$; **Infinity** (an inductive set exists); **Separation** (from a set, carve out by any formula — the safe replacement of naive comprehension); **Replacement** (images of sets under definable functions are sets); **Foundation** (no infinite $\in$ -chains; every nonempty set has an $\in$ -minimal element). **Choice** is *not* among them — it is the guest of §5.

Remark 4.3 (What ZF cannot do alone). ZF proves each $X_i \neq \emptyset$ has elements, but not that an *infinite family* admits a simultaneous choice: Russell’s socks (no distinguishing property) vs shoes (left/right gives a rule). Everything in §5 measures exactly this gap.

5 The Axiom of Choice and Its Equivalents

Definition 5.1 (AC, WO, Zorn). **AC**: every family $\{X_i\}_{i \in I}$ of nonempty sets has a choice function $c : I \rightarrow \bigcup_i X_i$ with $c(i) \in X_i$; equivalently $\prod_i X_i \neq \emptyset$. **WO (Well-Ordering Principle)**: every set admits a well-order. **Zorn (ZL)**: if $(P, \preceq) \neq \emptyset$ and every chain has an upper bound in P , then P has a maximal element.

Proposition 5.2 ($\text{WO} \Rightarrow \text{AC}$, in ZF).

Proof. Well-order $W = \bigcup_i X_i$. For each i , X_i is a nonempty subset of W , hence has a W -least element; set $c(i) = \min_W X_i$. This is a choice function — no choice used in the proof. $\square$

Proposition 5.3 ($\text{Zorn} \Rightarrow \text{AC}$, in ZF).

Proof. Let $\mathcal{F}$ be the set of partial choice functions: $f \in \mathcal{F}$ iff f is a function with $\text{dom}(f) \subseteq I$ and $f(i) \in X_i$. Order $\mathcal{F}$ by extension; $\mathcal{F} \neq \emptyset$ (empty function). A chain $\mathcal{C}$ has upper bound $\bigcup \mathcal{C}$ (union of compatible functions is a partial choice function). Zorn gives maximal f . If $\text{dom}(f) \neq I$, pick $i \notin \text{dom}(f)$ and $x \in X_i$ (a *single* choice, legal in ZF) and extend f by $i \mapsto x$ — contradicting maximality. Hence $\text{dom}(f) = I$. $\square$

Proposition 5.4 ($\text{AC} \Rightarrow \text{WO}$ and $\text{AC} \Rightarrow \text{Zorn}$: sketches). $\text{AC} \Rightarrow \text{WO}$: fix a choice function c on $\mathcal{P}(X) \setminus \{\emptyset\}$; by transfinite recursion set $x_\alpha = c(X \setminus \{x_\beta : \beta < \alpha\})$; the recursion must stop (no injection from the class of ordinals into a set — Hartogs), and at the stopping stage X is exhausted, so $\alpha \mapsto x_\alpha$ well-orders X . $\text{AC} \Rightarrow \text{Zorn}$: assuming no maximal element, recursively build a strictly increasing chain x_α by choosing at each stage a strict upper bound of $\{x_\beta : \beta < \alpha\}$ (which exists since chains have upper bounds and no upper bound is maximal); again Hartogs bounds the length — contradiction.

Corollary 5.5. Over ZF, the three statements AC, WO, ZL are equivalent (Propositions 5.2, 5.3, 5.4); so are Hausdorff’s Maximality Principle and Tukey’s lemma.

Theorem 5.6 (The list of equivalents and near-equivalents). Over ZF, the following are *equivalent* to AC:

- (1) Every vector space has a basis (Blass, 1984).
- (2) Tychonoff’s theorem: arbitrary products of compact spaces are compact.
- (3) Tarski’s theorem: for every infinite A , $|A \times A| = |A|$.
- (4) Every connected graph has a spanning tree (forward direction: Zorn on forests ordered by inclusion).

(5) Every surjection has a section (a reformulation of AC itself).

And the following live in strictly weaker tiers (important to know, easy to overstate): the **Boolean Prime Ideal / Ultrafilter Lemma** and the general **Gödel Completeness Theorem** (equivalent to BPI; the countable-language case is provable in ZF); **Hahn–Banach**; and mere consequences of AC such as the existence of non-measurable sets.

Remark 5.7 (Where the course spends Zorn). Krull’s theorem (S1: maximal ideals exist) $\rightarrow$ algebraic closures (S3) $\rightarrow$ Deligne’s theorem (S58: coherent topoi have enough points). Three existence theorems, one engine — and each is a “points exist” statement, fitting the course’s theme perfectly.

6 Exercises

Exercise 1. Prove both directions of Theorem 1.3 for: congruence mod n on $\mathbb{Z}$; and $x \sim y \iff |x| = |y|$ on $\mathbb{R}$ (describe the partition geometrically).

Exercise 2. Carry out the construction of $\mathbb{Q}$ from $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$: verify $(a, b) \sim (c, d) \iff ad = bc$ is an equivalence relation, and that $[(a, b)] + [(c, d)] = [(ad + bc, bd)]$ is well defined.

Exercise 3. Show $f \preceq g \iff f = O(g)$ is a preorder on positive functions $\mathbb{N} \rightarrow \mathbb{R}_{>0}$ but not a partial order; describe the quotient poset (growth rates) and identify its minimum.

Exercise 4. In $(\mathcal{P}(X) \setminus \{\emptyset\}, \subseteq)$ with $|X| \geq 2$: find minimal and maximal elements; does a minimum exist? Repeat for $(\mathbb{N}_{\geq 2}, |)$.

Exercise 5. Prove Proposition 2.2 in full (all case analyses), and show lexicographic order on $\mathbb{R}^2$ is a strict total order.

Exercise 6. Prove the forward direction of Proposition 3.4 in ZF, and pinpoint exactly where the converse uses dependent choice; give a poset with no decreasing chain that is not well-ordered.

Exercise 7. Prove Proposition 5.3 in full detail (verify $\bigcup \mathcal{C}$ is a function and a partial choice function; verify maximality forces full domain).

Exercise 8. Use Zorn to prove Krull’s theorem: every proper ideal of a commutative ring with $1 \neq 0$ lies in a maximal ideal (this is S1’s engine — prove it here first).

Exercise 9. Use Zorn to prove every vector space has a basis (order the linearly independent subsets by inclusion); identify where the chain upper bound uses unions.

Exercise 10. Prove Tychonoff’s theorem for *finite* products in ZF by induction, and locate the exact step where the general proof must invoke choice (ultrafilters or subbases).

Exercise 11. Show $AC \iff$ every surjection $p : E \rightarrow B$ has a section $s : B \rightarrow E$ ($p \circ s = \text{id}_B$); interpret sections as “continuous choices of points of fibers” and compare with Remark 1.4.

Exercise 12. Explain why Russell’s paradox does not contradict Separation: for a set A , show $\{x \in A : x \notin x\} \notin A$; conclude no universal set exists in ZF.

Where This Session Feeds the Course

- **Depends on:** nothing — this is session zero.
- **Feeds:** S1 (Krull via Zorn; quotients R/I); S2 (tensor as quotient; choosing representatives); S3 (algebraic closure via Zorn); S27 (sheafification classes); S34/S16 (orderings of finite subextensions for inverse limits); S58 (Deligne via Zorn); P1 (cosets as partitions, Exercise 12 preview).

- **Unifying thread:** quotient by equivalence = identifying points that must agree; order = the scaffolding on which Zorn climbs; Zorn = the engine that guarantees the *existence* of the points (maximal ideals, closures, models) the course studies.

References for this session

- [1] Halmos, P.R. *Naive Set Theory* (the classic first read).
- [2] Enderton, H.B. *Elements of Set Theory* (proofs of the equivalence cycle).
- [3] Jech, T. *The Axiom of Choice / Set Theory* (the definitive treatment).
- [4] Howard, P. & Rubin, J. *Consequences of the Axiom of Choice* (the list, with tiers).
- [5] Davey, B. & Priestley, H. *Introduction to Lattices and Order* (order theory).

End of Session P0 — next: Session P1, Group Theory: Part I